

Supporting Sciex ZT scanning-quadrupole DIA in Pioneer.jl

Pioneer.jl — branch `feat/sciex-zt-scanning` (off `perf/pep099-bitvec-random`)

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1 Summary

Sciex “ZT scanning” DIA (a.k.a. Scanning SWATH) sweeps a narrow-ish quadrupole transmission window of physical width W continuously across the precursor range, sampling as it moves. Groups of overlapping samples are merged on-instrument into “merged scans”; ProteoWizard then re-bins these into *contiguous* ≈ 1 m/z $Q1$ bins. We never see the true underlying scan — only the merged bins. A precursor at true m/z p therefore contributes ions to *several adjacent bins*, most to the bin centered nearest p and decaying $\approx$ linearly to either side. This document plans (1) a principled $mzML \rightarrow$ Pioneer conversion and (2) the search-side handling: run the fragment index at the *bin* tolerance, expand each precursor–scan hit to its neighboring bins (a “meta-scan”), deconvolve per bin, then reduce the meta-scan to a single row using the observed-vs-expected (triangular) weight profile as both a discriminative feature and an empirical calibration of the true effective window.

2 What the mzML actually contains (REP1, measured)

File `LFQ_ZenoTOF8600_ZTScanDIA_5Da_5min_50ng_Condition_A_REP1.mzML` (8.1 GB, ABI WIFF $\rightarrow$ `pwiz_Reader_ABI 3.0.21229`, centroided). Exhaustive inspection (full header, every per-spectrum `cvParam` and `userParam`, spectrum titles, `NativeIDs`, index):

- **The physical window $W = 5$ Da appears *nowhere* in the mzML** — not in `scanSettingsList` (absent), instrument config, per-scan params, titles, or index. It exists *only* in the filename (`..._5Da_...`). ProteoWizard discards the true $Q1$ width on conversion.
- **Slide/bin step $S = 1.0221$ m/z** — exact and uniform (495/495 consecutive center steps identical). Bins are *contiguous* (isolation lower/upper offset = 0.51, so edges abut). So the reported ≈ 1 m/z isolation width *is* the bin step S , not W .
- **Per-bin metadata present:** isolation target m/z (`MS:1000827`) = bin center; `preset scan configuration` = window index within a cycle; `NativeID cycle=C experiment=E`.
- $Q1$ center range 393.0–898.9 (span 506 m/z); **496 MS2 windows + 1 MS1 per cycle**; 818 cycles; 406,250 spectra; ≈ 2 ms/scan, ≈ 1 s/cycle over the 5-min gradient.

Consequence: W is the *only* quantity external to the file (a method constant); S , bin centers, cycle boundaries, and window order are all robustly recoverable. And — crucially — we do *not* need to trust $W = 5$: the empirical meta-scan weight profile of confident, high-abundance precursors (§8) reveals the true effective span directly.

3 Physical / signal model

Let bins be indexed b with centers $c_b = c_0 + bS$. A scanning window of width W moving in steps S transmits a precursor at p whenever the window overlaps p ; after re-binning, p 's ion current lands in a contiguous run of bins around $b^* = \arg \min_b |c_b - p|$. Define the meta-scan half-width

$$k = \text{round}\left(\frac{W - S}{S}\right) \Rightarrow \text{meta-scan} = \{b^* - k, \dots, b^* + k\}, \quad 2k + 1 \text{ bins.}$$

For $W = 5$, $S = 1.022$: $k = \text{round}(3.89) = 4$, i.e. a **9-bin meta-scan** (± 4), matching the user's $(W - S) \cdot 2 + 1 = 9$. The expected per-bin contribution is triangular,

$$t_j \propto \max(0, 1 - |j|/(k + 1)), \quad j = -k, \dots, k,$$

peaking at the center bin and decaying to (near) zero at $\pm k$. (The exact shape is the convolution of the true Q1 transmission profile with the bin/step kernel; a triangle is the first-order approximation and its *observed* form is what we calibrate against real data.)

4 Plan overview (pipeline stages)

1. **Conversion** — convert mzML preserving the ≈ 1 m/z bins; record W (filename/param), measure S , tag each MS2 scan with (cycle, window-index) so bins are addressable as a grid.
2. **Parameter tuning** — run with a *bin-width* precursor tolerance ($\approx S$): only the center bin, where the precursor contributes most, is used to fit mass error / iRT / NCE.
3. **Main search — fragment index** — run at the *bin* tolerance ($\approx S$), giving a list of (precursor_idx, scan_idx) on center bins only.
4. **Meta-scan expansion** — expand each hit to its $2k+1$ neighboring bins (same cycle): (pid, b^{*}) $\rightarrow$ (pid, b^{*}-k..b^{*}+k). Re-sort. The candidate list grows $\approx (2k+1) \times$.
5. **Deconvolution** — estimate per-bin weights/features, treating each bin as a *square* quad window of effective width $\approx (2k+1)S$ (defer true shape; §7).
6. **Triangular reduction** — for each precursor's $2k+1$ weights, compute observed-vs-triangle shape features and *collapse to the center row*, discarding the other $2k$.
7. **Scoring** — proceed as normal, with the new shape features available to the classifier.
8. **Chromatogram integration** — treat each $2k+1$ group as a meta-scan; the meta-scan weight is the triangle-weighted average of its bin weights; integrate the meta-scan-weight chromatogram.

5 Conversion (Routines/mzmlConverter/convertMzML.jl)

The converter already reads per-scan `centerMz` / `isolationWidthMz` and derives `cycle_idx` by detecting when the window returns to the first MS2 center (`CYCLE_IDX_ISOLATION_WINDOW_TOLERANCE_MZ`). ZT additions:

- **W input:** new conversion parameter `zt_window_width_mz`; convenience auto-parse of the `...<N>Da...` filename pattern as the default, overridable. Persist W (and measured S) in the arrow/run meta-data.

- **S measurement:** median of consecutive in-cycle `centerMz` steps (robust to the 1.02/1.022 alternation). Assert near-constant; warn otherwise.
- **Window index:** store `window_idx` (from `preset scan configuration` / center rank within a cycle) so `(cycle_idx, window_idx)` is a 2-D grid address for meta-scan neighbor lookup.
- **Keep bins as-is:** `centerMz` and `isolationWidthMz` $\approx S$ unchanged (do *not* bake W into `isolationWidthMz`); the W /meta-scan logic lives in the search so the per-bin structure and triangular weights are available.

Sanity: $818 \text{ cycles} \times 497 = 406,246$ ($\approx$ the 406,250 count; last cycle partial).

6 Search-side handling

Fragment index at bin tolerance. `get_frag_bounds.jl` builds candidate precursor m/z bounds from `centerMz` $\pm$ `isolationWidthMz`/2. With bins as-is this is the $\approx S$ center tolerance — exactly step (3). No change needed beyond feeding the bin width.

Meta-scan expansion (new step, post-frag-index / pre-deconvolution). Given the `(pid, scan)` hit list on center bins, for each hit add the $2k$ sibling bins at `window_idx` $\pm 1.. \pm k$ within the same `cycle_idx` (grid lookup). Guard edges (Q1 range 393–899) and missing bins. Re-sort by the deconvolution key (`scan`, then precursor). Track, per expanded row, its offset $j \in [-k, k]$ from the center bin and a `meta_group_id = (pid, cycle, b^*)`.

Deconvolution. Unchanged mechanics; each bin deconvolved with a square quad of width $\approx (2k+1)S$ (§7). Cost: $\approx (2k+1) \times$ more rows through deconvolution ($9 \times$ here) — the dominant new expense; see §9.

Triangular reduction (new step). For each `meta_group_id`, gather the $2k+1$ deconv weights $w_{-k}, \dots, w_k$ (ordered by j) and compute:

- `tri_dot` = $\langle \hat{w}, t \rangle$, normalized cosine of observed weights vs the ideal triangle t ;
- `tri_apex_offset` = $\arg \max_j w_j$ (should be ≈ 0 for a real precursor at the assumed center);
- `tri_symmetry`, `tri_center_fraction` = $w_0 / \sum_j w_j$, `tri_spread`;
- `center_weight` = w_0 (the retained quant).

Then **keep the center row** ($j = 0$), attach these features, and drop the other $2k$. A true precursor shows a centered, triangular profile; interference / wrong assignment shows a flat, shifted, or multi-modal profile — discriminative signal the classifier can use.

Chromatogram integration. In the integration search, deconvolve each bin of the meta-scan per cycle, then set the meta-scan weight to the *triangle-weighted average* $\bar{w} = \sum_j t_j w_j / \sum_j t_j$ (or the center weight; A/B). The integrated chromatogram is the time series of $\bar{w}$ over cycles. Reuse the same `meta_group/offset` bookkeeping.

7 Quadrupole model

The true transmission is a scanning convolution, so the per-bin quad shape is “cooked.” For now model each bin as a **perfectly square** isolation window of effective width $\approx (2k+1)S \approx 9\text{ m/z}$ (`SquareQuadModel`); skip `QuadTuningSearch` fitting for ZT runs (or force square). Refining to the real Q1 shape (estimating W and the transmission profile from data) is deferred; the triangular weight calibration (§8) is the first step toward it.

8 Diagnostics & empirical calibration of W

For a set of **confident, high-abundance** precursors (low q-value, high intensity), plot the observed meta-scan weight profile $w_{-k..k}$ against the expected triangle t . This directly reveals:

- the **true effective span** (where $w_j \rightarrow 0$) — if it is not ± 4 , W is not 5 and we adjust k empirically from the data rather than trusting the filename;
- the **true shape** (triangle vs rounded/trapezoidal) — informs the ideal t and, later, the quad model;
- **apex bias** (systematic `tri_apex_offset` $\neq 0$) — a Q1 calibration offset.

Deliverable: per-precursor observed-vs-expected overlays for a panel of good and bad PSMs (envigated diagnostic hook, like the existing chromatogram debug plots), plus an aggregate mean profile used to set k and t .

9 Risks / open questions

- $9\times$ **deconvolution cost** is the main expense; the meta-scan reduction happens *after* deconvolution, so peak RAM/time scale with the expanded list. Mitigate: expand only around frag-index hits (already sparse), reduce as early as possible, chunk by cycle.
- **Exact effective width & shape** — calibrate from §8; don’t hard-code $k = 4$ without checking REP1.
- **Sort/ordering invariants** — deconvolution and integration assume specific scan/precursor orderings; expansion must preserve them (re-sort + assertions).
- **iRT / cycle timing** — a meta-scan spans ~ 9 bins ($\sim 18\text{ ms}$) but one cycle; RT is the cycle time. Confirm chromatogram cycle indexing is by `cycle_idx`, not bin.
- **Interaction with the inherited scheme** — this branch carries PEP0.99 + random bitvec sampling + candidate-count logging; verify none interfere with the ZT path (bitvec calibration on $\approx 1\text{ m/z}$ bins in particular).

10 Phased implementation & tests

1. **Phase 0 — conversion + exploration.** Convert REP1 with `zt_window_width_mz`; verify S , cycle/window grid, bin counts. Produce the §8 weight-profile plots from a quick center-bin-only search to calibrate k/t . *No scoring changes yet.*
2. **Phase 1 — meta-scan expansion + square quad.** Expand hits $\pm k$, deconvolve, reduce to center row (no shape features yet). Verify IDs are non-zero and quant is sane vs the merged-bin naive run.

3. **Phase 2 — triangular features.** Add `tri_*` features + reduction; measure ID/FDR impact (entrapment).
4. **Phase 3 — meta-scan chromatogram integration.** Triangle-weighted meta-scan chromatograms; measure quant (CV, fold-change).
5. **Phase 4 — quad-shape refinement (optional).** Fit the real effective width/shape from the calibration data.

Success criteria: REP1 yields a sensible number of confident IDs at 1% FDR with a centered/-triangular mean weight profile; entrapment FDR controlled; quant CV comparable to standard DIA. Data: `RIS:NTW/SciexZT_ZTScanDIA` (REP1–3); library = the 3P Sciex library used for the nSWATH bench data (`3P_May7-2026_bitvec_10.poin`) or a matched human library — TBD.